Assignment 11

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Problem 1

Use the diabetes data (copy and paste the following command into R console:library(faraway);data("diabetes")).

Four hundred three African Americans were interviewed in a study to understand the prevalence of obesity, diabetes and other cardiovascular risk factors in central Virginia. Data is presented in diabetes. In this question we are interested in building a model to predict glycosolated hemoglobin (glyhb) in terms of the other relevant variables. (40 points)

a-) use set.seed(1023), to create test(30% of the observations) and train (70% of the observations) data set. (5 points)

Since there are many missing values in the *diabetes* dataset, we first eliminate NAs by dropping two columns, *bp.2s* and *bp.2d* because these two columns have too many NAs. Next, we delete all the rows that have NAs. *id* column is also dropped because there is no meaning. Lastly, we create dummy variables for factor variables since 'neuralnet' function only works with quantitative variables. It makes sense because we will have to normalize the data.

```
data(diabetes)
str(diabetes)
```

```
## 'data.frame':    403 obs. of  19 variables:
##  $ id      : 'labelled' int  1000 1001 1002 1003 1005 1008 1011 1015 1016 1022 ...
##   ..- attr(*, "label")= chr "Subject ID"
##  $ chol    : 'labelled' int  203 165 228 78 249 248 195 227 177 263 ...
##   ..- attr(*, "label")= chr "Total Cholesterol"
##  $ stab.glu: 'labelled' int  82 97 92 93 90 94 92 75 87 89 ...
##   ..- attr(*, "label")= chr "Stabilized Glucose"
##  $ hdl     : 'labelled' int  56 24 37 12 28 69 41 44 49 40 ...
##   ..- attr(*, "label")= chr "High Density Lipoprotein"
##  $ ratio   : 'labelled' num  3.6 6.9 6.2 6.5 8.9 ...
##   ..- attr(*, "label")= chr "Cholesterol/HDL Ratio"
##  $ glyhb   : 'labelled' num  4.31 4.44 4.64 4.63 7.72 ...
##   ..- attr(*, "label")= chr "Glycosolated Hemoglobin"
##  $ location: Factor w/ 2 levels "Buckingham","Louisa": 1 1 1 1 1 1 1 1 1 1 ...
##  $ age     : int  46 29 58 67 64 34 30 37 45 55 ...
##   ..- attr(*, "units")= chr "years"
##  $ gender  : Factor w/ 2 levels "male","female": 2 2 2 1 1 1 1 1 1 2 ...
##  $ height  : int  62 64 61 67 68 71 69 59 69 63 ...
##   ..- attr(*, "units")= chr "inches"
##  $ weight  : int  121 218 256 119 183 190 191 170 166 202 ...
##   ..- attr(*, "units")= chr "pounds"
##  $ frame   : Factor w/ 3 levels "small","medium",..: 2 3 3 3 2 3 2 2 3 1 ...
##  $ bp.1s   : 'labelled' int  118 112 190 110 138 132 161 NA 160 108 ...
##   ..- attr(*, "label")= chr "First Systolic Blood Pressure"
```

```
##  $ bp.1d   : 'labelled' int  59 68 92 50 80 86 112 NA 80 72 ...
##   ..- attr(*, "label")= chr "First Diastolic Blood Pressure"
##  $ bp.2s   : 'labelled' int  NA NA 185 NA NA NA 161 NA 128 NA ...
##   ..- attr(*, "label")= chr "Second Systolic Blood Pressure"
##   ..- attr(*, "comment")= chr "equals first measurement if it was not high"
##  $ bp.2d   : 'labelled' int  NA NA 92 NA NA NA 112 NA 86 NA ...
##   ..- attr(*, "comment")= chr "equals first measurement if it was not high"
##   ..- attr(*, "label")= chr "Second Diastolic Blood Pressure"
##  $ waist   : int  29 46 49 33 44 36 46 34 34 45 ...
##   ..- attr(*, "units")= chr "inches"
##  $ hip     : int  38 48 57 38 41 42 49 39 40 50 ...
##   ..- attr(*, "units")= chr "inches"
##  $ time.ppn: 'labelled' int  720 360 180 480 300 195 720 1020 300 240 ...
##   ..- attr(*, "label")= chr "Postprandial Time when Labs were Drawn"
##   ..- attr(*, "units")= chr "minutes"
```

```
head(diabetes[,1:12])
```

id	chol	stab.glu	hdl	ratio	glyhb	location	age	gender	height	weight	frame
1000	203	82	56	3.6	4.31	Buckingham	46	female	62	121	medium
1001	165	97	24	6.9	4.44	Buckingham	29	female	64	218	large
1002	228	92	37	6.2	4.64	Buckingham	58	female	61	256	large
1003	78	93	12	6.5	4.63	Buckingham	67	male	67	119	large
1005	249	90	28	8.9	7.72	Buckingham	64	male	68	183	medium
1008	248	94	69	3.6	4.81	Buckingham	34	male	71	190	large

```
head(diabetes[,13:19])
```

bp.1s	bp.1d	bp.2s	bp.2d	waist	hip	time.ppn
118	59	NA	NA	29	38	720
112	68	NA	NA	46	48	360
190	92	185	92	49	57	180
110	50	NA	NA	33	38	480
138	80	NA	NA	44	41	300
132	86	NA	NA	36	42	195

```
apply(diabetes,2,function(x) sum(is.na(x))) # check missing values
```

```
##       id     chol stab.glu      hdl    ratio    glyhb location      age
##        0        1        0        1        1       13        0        0
##   gender   height   weight    frame    bp.1s    bp.1d    bp.2s    bp.2d
##        0        5        1       12        5        5      262      262
##    waist      hip time.ppn
##        2        2        3
```

```
diabetes2 <- diabetes[,-c(1,15,16)] # drop id, bp.2s, bp.2d
diabetes2 <- na.omit(diabetes2) # delete rows containing NAs
X <- model.matrix(glyhb~., data=diabetes2)[,-1]
diabetes3 <- data.frame(cbind(glyhb=diabetes2$glyhb,X))

set.seed(1023)
train_index <- createDataPartition(y=diabetes3$glyhb, p=0.7, list=FALSE)
diabetes_train <- diabetes3[train_index,]
diabetes_test <- diabetes3[-train_index,]
```

Use the train data set to answer part b and c below:

b-) Compute the total sum of squares for the response, glyhb. This is the sum of squares about the mean.

Now fit a neural network model with this response and the remaining variables as predictors. Use two hidden nodes. Do not rescale and fit the model only once. Compute the residual sum of squares. How has the model performed? (10 points)

```
sst <- sum((diabetes_train$glyhb - mean(diabetes_train$glyhb))^2)

set.seed(1023)
NN <- neuralnet(glyhb ~ ., hidden=2, data=diabetes_train, linear.output=T)
plot(NN)
pred_NN <- compute(NN, diabetes_train[,-1])
sse_NN_train <- sum((pred_NN$net.result - diabetes_train$glyhb)^2)
sse_NN_train
```

```
## [1] 1328.747
```

Total Sum of squares (SST): 1328.747
Residual sum of squares (SSE): 1328.747

SSE is equal to SST, so the model is no better than taking the mean as its prediction.

c-) Rescale all the numerical variables in the data frame and recompute the total sum of squares. Refit the same neural network model but to the rescaled data frame. Has the model done any better? (10 points)

```
# Rescale all variables in between 0 and 1 scale
normalize <- function(x) {return((x - min(x)) / (max(x) - min(x)))}
scaled_diabetes <- as.data.frame(lapply(diabetes3, normalize))
scaled_dbt_train <- scaled_diabetes[train_index,]
scaled_dbt_test <- scaled_diabetes[-train_index,]

set.seed(1023)
NN2 <- neuralnet(glyhb ~ ., hidden=2, data=scaled_dbt_train, linear.output=T)
plot(NN2)
pred_NN2 <- compute(NN2, scaled_dbt_train[,-1])

unnormalize <- function(x) {return(x * (max(diabetes3$glyhb) - min(diabetes3$glyhb)) +
                                   min(diabetes3$glyhb))}
pred_NN2_un <- unnormalize(pred_NN2$net.result)
sse_NN2_train <- sum((pred_NN2_un - diabetes_train$glyhb)^2)
sse_NN2_train
```

```
## [1] 276.7339
```

After rescaling, the model performs better, and it is shown by the reduced SSE of 276.734.

- There were two different methods of normalizing, one from the lecture R file and another from the HW10 solutions. I wasn't sure which one is correct or whether both are correct, so I followed the one from the lecture R file.

d-) Test the model performances of both models on the test data sets. Calculate SSE (hint: for the model built in part C, you need to transform them back to original scale) (10 points)

```
pred_NN_test <- compute(NN, diabetes_test[,-1])
sse_NN_test <- sum((pred_NN_test$net.result - diabetes_test$glyhb)^2)

pred_NN2_test <- compute(NN2, scaled_dbt_test[,-1])
pred_NN2_test_un <- unnormalize(pred_NN2_test$net.result)
sse_NN2_test <- sum((pred_NN2_test_un - diabetes_test$glyhb)^2)

tbl <- rbind(sse_NN_test, sse_NN2_test)
```

```r
rownames(tbl) <- c("Unscaled Neural Network","Scaled Neural Network")
colnames(tbl) <- c("SSE")
kable(tbl, digits=4, align="c", booktabs=TRUE,caption="Model Comparison") %>%
  kable_styling(latex_options="HOLD_position")
```

Table 1: Model Comparison

	SSE
Unscaled Neural Network	488.1155
Scaled Neural Network	378.4994

e-) Which model would you pick? (5 points)

Based on the model comparison table provided in part(d), we would choose the neural network model fitted with the scaled data since it has a smaller SSE. This is the expected result as the neural network learns and converges faster with scaled data.

Problem 2

Refer to the Disease outbreak data set. Savings account status is the response variable and age, socioeconomic status, and city sector are the predictor variables. (30 points, 10 points each)

a-) Fit logistic regression model to predict the saving account status on the predictor variables in first-order terms and interaction terms for. all pairs of predictor variables. State the fitted response function.

All original column names are simplified as Y, X_1, X_2, X_3, X_4. Y is a dichotomous response variable, X_2 has 3 levels, X_3, X_4 have 2 levels, but X_3 is coded 1-2. We will create two dummy variables for X_2 and change the levels of X_3 to 0-1.

```r
disout.o <- getDataSet("Disease Outbreak.csv")
str(disout.o)
```

```
## 'data.frame':    196 obs. of  5 variables:
##  $ Age                   : int  33 35 6 60 18 26 6 31 26 37 ...
##  $ Socioeconomic.status  : int  1 1 1 1 3 3 3 2 2 2 ...
##  $ Sector                : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Disease.status        : int  0 0 0 0 1 0 0 1 1 0 ...
##  $ Savings.account.status: int  1 1 0 1 0 0 0 1 0 0 ...
```

```r
head(disout.o)
```

Age	Socioeconomic.status	Sector	Disease.status	Savings.account.status
33	1	1	0	1
35	1	1	0	1
6	1	1	0	0
60	1	1	0	1
18	3	1	1	0
26	3	1	0	0

```r
apply(disout.o,2,function(x) sum(is.na(x))) # check any missing data
```

```
##                     Age   Socioeconomic.status                 Sector
##                       0                      0                      0
##         Disease.status Savings.account.status
##                       0                      0
```

```r
# Simplify variable names
Y <- disout.o$Savings.account.status
X1 <- disout.o$Age
X2 <- disout.o$Socioeconomic.status
X3 <- disout.o$Sector
X4 <- disout.o$Disease.status
disout.pre <- data.frame(cbind(Y,X1,X2,X3,X4))
table(disout.pre$X2)
```

1	2	3
77	49	70

```r
table(disout.pre$X3)
```

1	2
117	79

```r
table(disout.pre$X4)
```

0	1
139	57

```r
disout <- dummy_cols(disout.pre, select_columns = 'X2')
disout <- dummy_cols(disout, select_columns = 'X3')
disout <- disout[,-c(3,4,6,9)] # drop unnecessary columns
head(disout)
```

Y	X1	X4	X2_2	X2_3	X3_2
1	33	0	0	0	0
1	35	0	0	0	0
0	6	0	0	0	0
1	60	0	0	0	0
0	18	1	0	1	0
0	26	0	0	1	0

```r
f1 <- glm(Y ~ X1+X4+X2_2+X2_3+X3_2+X1:X4+X1:X2_2+X1:X2_3+X1:X3_2+X4:X2_2+X4:X2_3+X4:X3_2+
            X2_2:X3_2+X2_3:X3_2, family = binomial, data=disout)
summary(f1)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X4 + X2_2 + X2_3 + X3_2 + X1:X4 + X1:X2_2 +
##     X1:X2_3 + X1:X3_2 + X4:X2_2 + X4:X2_3 + X4:X3_2 + X2_2:X3_2 +
##     X2_3:X3_2, family = binomial, data = disout)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.3855  -0.8886   0.4118   0.7943   2.0273
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.155908   0.587157   0.266  0.79060
## X1           0.035838   0.021966   1.632  0.10277
## X4          -0.946814   1.062247  -0.891  0.37275
## X2_2        -1.306280   0.817101  -1.599  0.10989
## X2_3        -2.151271   0.758426  -2.836  0.00456 **
## X3_2         0.916937   0.781780   1.173  0.24084
## X1:X4        0.021247   0.025814   0.823  0.41045
```

```
## X1:X2_2      0.008166   0.029619   0.276  0.78278
## X1:X2_3      0.002890   0.024113   0.120  0.90461
## X1:X3_2     -0.021077   0.022438  -0.939  0.34755
## X4:X2_2     -0.111640   1.044638  -0.107  0.91489
## X4:X2_3     -0.137603   0.958732  -0.144  0.88587
## X4:X3_2      0.930980   0.835249   1.115  0.26502
## X2_2:X3_2   -0.131848   0.880545  -0.150  0.88097
## X2_3:X3_2    0.388653   0.867955   0.448  0.65431
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 270.06  on 195  degrees of freedom
## Residual deviance: 212.84  on 181  degrees of freedom
## AIC: 242.84
##
## Number of Fisher Scoring iterations: 5
```

Y = 0.155908 + 0.0358383X1 - 0.946814X4 - 1.30628X2_2 - 2.15127X2_3 + 0.916937X3_2 + 0.0212472X1X4 + 0.00816572X1X2_2 + 0.00288974X1X2_3 - 0.0210772X1X3_2 - 0.11164X4X2_2 - 0.137603X4X2_3 + 0.93098X4X3_2 - 0.131848X2_2X3_2 + 0.388653X2_3X3_2

b-) Use the likelihood ratio test to determine whether all interaction terms can be dropped from the regression model; use $\alpha = .01$. State the alternatives, full and reduced models, decision rule, and conclusion. What is the approximate P-value of the test?

```
f1_reduced <- glm(Y ~ X1+X4+X2_2+X2_3+X3_2, family = binomial, data=disout)
anova(f1_reduced, f1, test="Chi")
```

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
190	215.3575	NA	NA	NA
181	212.8362	9	2.52127	0.980306

H_0: All interaction terms can be dropped (all of their coefficients are zero).
H_a: Not all interaction terms can be dropped (not all of their coefficients are zero).

The full and reduced models are described in ANOVA table as Model 1 and Model 2. ANOVA table above shows that p-value = 0.9803, which is greater than the significance level, α=0.01, so we accept H_0. All interaction terms can be dropped.

c-) Conduct the Hosmer-Lemeshow goodness of fit test for the appropriateness of the logistic regression function by forming five groups of approximately 20 cases each; use $\alpha = .05$.

```
f1.2 <- step(f1_reduced, direction="backward", trace=0)
summary(f1.2)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X2_2 + X2_3 + X3_2, family = binomial,
##     data = disout)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.2898  -0.8648   0.3887   0.8149   1.9887
##
## Coefficients:
##              Estimate Std. Error z value   Pr(>|z|)
```

```
## (Intercept)  0.054054   0.381415   0.142   0.887302
## X1           0.035471   0.009796   3.621   0.000294 ***
## X2_2        -1.174332   0.417764  -2.811   0.004939 **
## X2_3        -1.953575   0.402550  -4.853 0.00000122 ***
## X3_2         0.789524   0.348572   2.265   0.023511 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##     Null deviance: 270.06  on 195  degrees of freedom
## Residual deviance: 215.36  on 191  degrees of freedom
## AIC: 225.36
##
## Number of Fisher Scoring iterations: 4
```

```
hoslem.test(f1.2$y, fitted(f1.2), g=5)
```

```
##
##  Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  f1.2$y, fitted(f1.2)
## X-squared = 3.8928, df = 3, p-value = 0.2733
```

```
hoslem.test(f1_reduced$y, fitted(f1_reduced), g=5)
```

```
##
##  Hosmer and Lemeshow goodness of fit (GOF) test
##
## data:  f1_reduced$y, fitted(f1_reduced)
## X-squared = 3.8719, df = 3, p-value = 0.2756
```

H_0: The model is a good fit H_a: The model is not a good fit.

We used the reduced model to conduct the test since the likelihood ratio test indicated that we should drop all the interaction terms. Then, we checked if any of the first order terms can be dropped, and the backward elimination resulted in dropping X_4. The Hosmer-Lemeshow test shows p-value = 0.2733, which is greater than α=0.05, so we accept the null hypothesis. The model looks like a good fit. The test result of the reduced model before dropping X_4 is also shown, and it also looks like a good fit. (The question asks us to form 5 groups of about 20 cases each, but since we have 196 cases, forming 5 groups resulted in about 40 cases each. Forming 10 groups of about 20 cases each and conducting the test gives the same result of a good fit with bigger p-value.)

Problem 3

Refer to the Geriatric study. A researcher in geriatrics designed a prospective study to investigate the effects of two interventions on the frequency of falls. One hundred subjects were randomly assigned to one of the two interventions: education only ($X_1 = 0$) and education plus aerobic exercise training ($X_1 = 1$). Subjects were at least 65 years of age and in reasonably good health. Three variables considered to be important as control variables were gender (X_2:0=female;1=male), a balance index (X_3). and a strength index (X_4). The higher balance index, the more stableis the subject and the higher the strength index, the stronger is the subject. Each subject kept a diary recording the number of falls (Y) during the six months of the study.(30 points, 10 points each)

a-) Fit the regression model. State the estimated regression coefficients, their estimated standard deviations. and the estimated response function.

```r
geri <- getDataSet("Geriatric Study.csv")
str(geri)
```

```
## 'data.frame':    100 obs. of  5 variables:
##  $ Y : int  1 1 2 0 2 1 0 0 2 2 ...
##  $ X1: int  1 1 1 1 1 1 1 1 1 1 ...
##  $ X2: int  0 0 1 1 0 1 1 0 1 1 ...
##  $ X3: int  45 62 43 76 51 73 40 66 80 56 ...
##  $ X4: int  70 66 64 48 72 39 54 37 81 60 ...
```

```r
head(geri)
```

Y	X1	X2	X3	X4
1	1	0	45	70
1	1	0	62	66
2	1	1	43	64
0	1	1	76	48
2	1	0	51	72
1	1	1	73	39

```r
apply(geri,2,function(x) sum(is.na(x)))
```

```
##  Y X1 X2 X3 X4
##  0  0  0  0  0
```

```r
f3 <- glm(Y ~ X1+X2+X3+X4, family=poisson, data=geri)
summary(f3)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X2 + X3 + X4, family = poisson, data = geri)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.1854  -0.7819  -0.2564   0.5449   2.3626
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.489467   0.336869   1.453  0.14623
## X1          -1.069403   0.133154  -8.031 9.64e-16 ***
## X2          -0.046606   0.119970  -0.388  0.69766
## X3           0.009470   0.002953   3.207  0.00134 **
## X4           0.008566   0.004312   1.986  0.04698 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 199.19  on 99  degrees of freedom
## Residual deviance: 108.79  on 95  degrees of freedom
## AIC: 377.29
##
## Number of Fisher Scoring iterations: 5
```

```r
tbl2 <- data.frame(summary(f3)$coefficients[,1:2])
rownames(tbl2) <- c("b0","b1","b2","b3","b4")
```

```r
colnames(tbl2) <- c("Coefficient Estimates","Standard Deviations")
kable(tbl2, digits=4, align="c", booktabs=TRUE,
      caption="Estimated Regression Coefficients and SD") %>%
  kable_styling(latex_options="HOLD_position")
```

Table 2: Estimated Regression Coefficients and SD

	Coefficient Estimates	Standard Deviations
b0	0.4895	0.3369
b1	-1.0694	0.1332
b2	-0.0466	0.1200
b3	0.0095	0.0030
b4	0.0086	0.0043

The estimated response function:
$\hat{\mu} = e^{0.4895-1.0694X_1-0.0466X_2+0.0095X_3+0.0086X_4}$

b-) Assuming that the fitted model is appropriate, use the likelihood ratio test to determine whether gender (X_2) can be dropped from the model: State the full and reduced models. decision rule. and conclusion. What is the P-value of the test

```r
f3_reduced <- glm(Y ~ X1+X3+X4, family=poisson, data=geri)
summary(f3_reduced)
```

```
##
## Call:
## glm(formula = Y ~ X1 + X3 + X4, family = poisson, data = geri)
##
## Deviance Residuals:
##     Min       1Q   Median       3Q      Max
## -2.2152  -0.7512  -0.2594   0.5830   2.2893
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.443890   0.317289   1.399  0.16181
## X1          -1.077770   0.131415  -8.201 2.38e-16 ***
## X3           0.009471   0.002957   3.203  0.00136 **
## X4           0.008979   0.004190   2.143  0.03209 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##     Null deviance: 199.19  on 99  degrees of freedom
## Residual deviance: 108.94  on 96  degrees of freedom
## AIC: 375.44
##
## Number of Fisher Scoring iterations: 5
```

```r
anova(f3_reduced, f3, test="Chisq")
```

Resid. Df	Resid. Dev	Df	Deviance	Pr(>Chi)
96	108.9409	NA	NA	NA
95	108.7899	1	0.1509982	0.6975832

H_0: X_2 can be dropped.
H_a: X_2 cannot be dropped.

The reduced and full model are stated in ANOVA table as Model 1 and Model 2. The p-value = 0.6976, which is greater than our significance level, $\alpha = 0.05$, so we accept the null hypothesis that X_2 can be dropped.

c-) Predicted the number of falls for X_1=1, X_2=0, X_3=45, X_4=70

```
test_dat <- data.frame(X1=1, X2=0, X3=45, X4=70)
pred3 <- predict(f3_reduced, test_dat, type="link")
exp(pred3)
```

```
##       1
## 1.52328
```

We used the reduced model from part(b) since the likelihood ratio test showed that we should drop X_2. The predicted number of falls for X_1=1, X_2=0, X_3=45, X_4=70 is 1.5233.